

BASTAV PODDER

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EDUCATION

NIT Durgapur

Civil Engineering - **CGPA - 7.57**

08 2023 – 06 2027

Kolkata, India

ACME Academy

ICSE **Percentage - 85.7%**

04 2021 – 04 2023

Kolkata, India

COURSEWORK / SKILLS

- Data Structures & Algorithms
- Cloud Architecture
- Database Management System (DBMS)
- Computer Networks
- Operating System
- High Level Designs
- OOPS Concepts
- Web Development

PROJECTS

Streamly | [Next.js 14](#), [Stream SDK](#), [WebRTC](#), [Clerk](#), [TypeScript](#), [Tailwind](#)

2025

- Built a production-grade, full-stack video conferencing platform using Next.js 14 App Router, Stream Video SDK, and WebRTC, enabling real-time multi-participant calls with **sub-200ms connection latency**.
- Integrated Clerk for full-cycle authentication and middleware-level route protection, reducing authentication setup complexity by **60%** and ensuring 0% unauthorized route access.
- Architected a meeting scheduling system using Next.js Server Actions and Stream's API, supporting instant and future-dated sessions via **UUID-based shareable links**.
- Developed a multi-view dashboard with client-side state managed through custom React hooks, reducing redundant server round-trips by **35%** and improving UI responsiveness.
- Structured a modular, strictly-typed TypeScript codebase across actions, hooks, and providers, improving developer velocity by **45%** through compile-time interface safety.

Redis-Clone | [C++17](#), [Sockets](#), [Multithreading](#), [STL](#), [RESP Protocol](#), [AOF Persistence](#)

2026

- Architected a multithreaded, Redis-compatible in-memory data store in C++17, implementing 3 core data structures (strings, lists, hashes) and over 25 client commands, using a thread-per-connection concurrency model to handle simultaneous client connections.
- Ensured thread-safe, multi-user data access via fine-grained mutex synchronization, using a recursive locking strategy around all shared in-memory stores to guarantee data consistency and prevent race conditions when multiple clients read/write concurrently.
- Designed and implemented a custom RESP (REdis Serialization Protocol) parser capable of handling fragmented TCP streams and pipelined command batches, enabling reliable communication with standard Redis-compatible clients without reliance on third-party networking libraries.
- Engineered a dual-layer persistence architecture combining Append-Only File (AOF) logging for command-level durability with periodic full-state snapshotting every 300 seconds, plus graceful-shutdown handling that automatically persists the entire in-memory dataset to disk on termination (e.g., SIGINT/SIGTERM) to guarantee zero data loss across restarts.
- Developed a thread-safe Publish/Subscribe messaging subsystem supporting real-time, low-latency message broadcast to multiple concurrently connected clients, complemented by a real-time monitoring dashboard for live server metrics and an automated test suite validating core and extended functionality.

TECHNICAL SKILLS

Languages: C, C++, JavaScript, TypeScript, Python, MySQL

Developer Tools: Docker, Jenkins, AWS, Git, GitHub, Postman

Technologies & Frameworks: ReactJS, Next.js, CI/CD, Containerization, Kubernetes, Docker, Node.js, Express.js, MongoDB, Tailwind CSS, REST APIs, WebRTC, Stream SDK, Clerk, Inngest, Nodemailer

ACHIEVEMENTS

Competitive Programming: Achieved a peak Contest Rating of **1758** on Leetcode.

Problem Solving: Solved **1250+** algorithmic problems across major platforms including **LeetCode(800+)** and **Codeforces(310+)**.

Global Recognition: Secured a Global Rank of **1187** out of 35000+ of participants in the LeetCode Weekly Contest 505.

CERTIFICATIONS

- C++ - Udemy.
- Web Development& Devops- 100xDevs.